Fiscal Year:	FY 2016	**Task Last Updated:**	FY 12/22/2015
PI Name:	Olson, Sandra Ph.D.		
Project Title:	Correlation of Normal Gravity Aerospace Materials Flammability Data with Flammability in Microgravity Environments		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	44135	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Combustion Science NNH09ZTT001N
Start Date:	01/01/2010	**End Date:**	12/31/2019
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Urban, David	**Contact Phone:**	216-433-2835
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Flight Program:			
Flight Assignment:	NOTE: Extended to 12/31/2019, per PI (Ed., 1/26/2015) NOTE: Extended to 12/31/2014, per PI (Ed., 10/17/13) NOTE: Extended to 12/31/2013, per PI (Ed., 10/10/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hirsch, David (NTEC, Jacobs Engineering)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NASA STD 6001 Test 1 is the major method used to evaluate flammability of materials intended for use in habitable environments of U.S. spacecraft. The method is an upward flame propagation test initiated in a stagnant environment and using a well-defined igniter flame at the bottom of a vertically mounted sample. A material passes this test if the vertical burn length is less than 15.2 cm (6 inches) and there is no evidence of transfer of burning debris. The test is conducted in the most severe flaming combustion environment (oxygen concentration, pressure) expected in the spacecraft, currently 30% oxygen at 10.2 psia. Recent research has shown that current normal gravity materials flammability tests do not correlate with flammability in ventilated, micro-, or reduced-gravity conditions. The materials selection for spacecraft is based on the assumption of commonality between ground test flammability results and spacecraft environments, which does not appear to be valid. To better understand the actual normal gravity flammability limits, a modified Test 1 protocol has been proposed that, as an alternative to qualifying materials as pass/fail in the worst-expected environments, measures the actual upward flammability limit (akin to the Limiting Oxygen Index (LOI)) for the material so that a more accurate assessment of the margin of safety of the material in the real use environment can be made. The method allows the option of selecting better or best materials based on the margin of safety, as opposed to what would be considered just "passing" from a flammability point of view. A research program is underway to correlate normal gravity flammability test data with data under ventilated reduced gravity or microgravity conditions. The program will determine materials suitability for microgravity and reduced gravity testing by evaluating oxygen concentration thresholds under various total pressures in normal gravity tests and in spacecraft-specific environments (ventilated microgravity or reduced partial gravity). The normal gravity data will be correlated with flammability data in microgravity environments and the safety factors involved when selecting spacecraft materials based on normal gravity testing will be evaluated. Rigorous correlations between ground test flammability data and data in ventilated spacecraft environments would allow selection of materials with increased fire resistance, and thus decrease the fire risk in space systems. The operational flexibility will also increase since the safety factors will be assessable.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The identification of the fundamental limit in microgravity and its comparison with the normal gravity flammability limit will help us understand flammability in different configurations on Earth. It will improve the safety of astronauts in spaceflight so they can return safely to Earth.
Task Progress:	Microgravity experiments burning cast Polymethylmethacrylate (PMMA) cylindrical rods in axial stagnation flow were conducted aboard the International Space Station in 2014 as part of the BASS-II (Burning and Suppression of Solids) experiment. The complete lower portion of the oxygen-stretch rate flammability boundary was obtained experimentally for PMMA in microgravity for the first time. Complementary 5.18-second Zero Gravity Facility blowoff drop tests were also conducted. A constant critical Damkohler number analysis has been applied to the data. For the blowoff branch, the boundary exhibits a linear dependence between oxygen concentration and stretch rate. For the quenching branch, the dominance of the exponential kinetics term indicates that the temperature is critical. A Nusselt number heat transfer correlation has been used to estimate the heat flux to the rod along the flammability boundary. Oxygen consumption calorimetry is then used to estimate the flow radius needed to provide adequate oxygen. The thermal length, oxygen flow length, and measured standoff distance are all within 1 mm across the blowoff region, but the thermal and oxygen lengths diverge once flows drop to the order of diffusive velocities with the flame standoff distance between them. A two zone surface energy balance uses the heat flux correlation and gas phase radiative feedback to evaluate relative importance of heat losses along the flammability boundary. At the bottom of the flammability boundary, heat losses amount to approximately 20% of the net heat flux to the surface. Quantitatively, heat losses are actually lowest along the quenching branch, but because of the low total flame heat flux, surface radiative loss becomes a dominant factor. Gas phase radiative feedback increases from 3% of the total heat flux at high stretch to 26% at low stretch. Measured rod regression rates agree with predicted values. A one-dimensional version of the model agrees with previously published stagnation flow regression rates. In order to more accurately determine the relationship between the burning of solid materials under microgravity and normal gravity conditions, a series of normal gravity blowoff extinction tests were conducted to compare with BASS-II testing. This testing utilized the BASS-II concurrent rod geometry, with the addition of normal gravity buoyant flow. Polymethylmethacrylate (PMMA) rods of diameters ranging from 0.25” to 1.5” were tested at oxygen percentages ranging from 14-18% by volume. The forced flow where blowoff occurred was found for each rod size and oxygen concentration. These blowoff flammability limit results are currently being evaluated. The blowoff flammability boundary when pressure and oxygen are varied instead of the forced flow was also investigated. The partial pressure of oxygen at blowoff increases linearly with total pressure, in a trend different from the simple volume fraction value. Based on these results, it seems plausible that an improved test method that determines the normal gravity blowoff boundary can be used to extrapolate to the reduced gravity material flammability boundary, which will improve spacecraft fire safety for future exploration missions.
Bibliography Type:	Description: (Last Updated: 01/12/2026)
Abstracts for Journals and Proceedings	Olson SL, Ferkul PV. "Critical Damkohler Number Flammability Boundary Analysis for PMMA Cylinders Burning in Axial Forced Concurrent Flow." 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. https://www.asgsr.org/ , Nov-2015

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Papers from Meeting Proceedings	Olson SL, Ferkul PV, Bhattacharjee S, Miller FJ, Fernandez-Pello C, Link S, T'ien JS, Wichman IS. "Results from on-board CSA-CP and CDM Sensor Readings during the Burning and Suppression of Solids – II (BASS-II) Experiment in the Microgravity Science Glovebox (MSG)." ICES-2015-196. 45th International Conference on Environmental Systems, Bellevue, Washington, July 12-16, 2015. 45th International Conference on Environmental Systems, Bellevue, Washington, July 12-16, 2015. Paper number ICES-2015-196. http://www.depts.ttu.edu/ceweb/ices/ , Jul-2015